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UNCONSTRAINED OPTIMIZATION I

Gradient Descent: Walking Downhill, One Step at a Time



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Three Questions, That Is the Whole Lecture

1

Which way is down?

Look at the slope under your feet and go the other way.

2

How big a step?

Too small: you crawl. Too big: you bounce. There is a sweet spot.

3

When do I stop?

When the ground is flat, or progress becomes tiny.

Plus two practical tricks: rescale your data, and always check your slope.

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Lost on a Mountain in Thick Fog

THE SITUATION

You want to reach the lowest point of a valley. The fog is so thick you can only see the ground under your feet.

THE STRATEGY

Feel which way the ground tilts. Take a step downhill. Feel again. Repeat until the ground is flat.

The dictionary

On the mountain	In optimization
your position	**the values x we are choosing**
height	**the error / cost f(x)**
tilt of the ground	**the slope (gradient)**
length of your stride	**the step size α**
the valley floor	**the best x (minimum)**

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The Slope Tells You Which Way Is Down

Our example for the whole lecture: **$f(x) = (x - 3)^2$** — a bowl with its bottom at x = 3. Its slope is **$2 \cdot (x - 3)$**

where you stand	slope	what it means	so you move
x = 7	+8	uphill to the right, steep	**LEFT, a lot**
x = 4	+2	uphill to the right, gentle	**LEFT, a little**
x = 3	**0**	**flat — the bottom!**	**STOP**
x = 1	−4	uphill to the left	**RIGHT**

THE RULE

Go AGAINST the slope.

Big slope → big move.
Zero slope → you are there.

The sign of the slope says WHICH WAY. The size of the slope says HOW STEEP.

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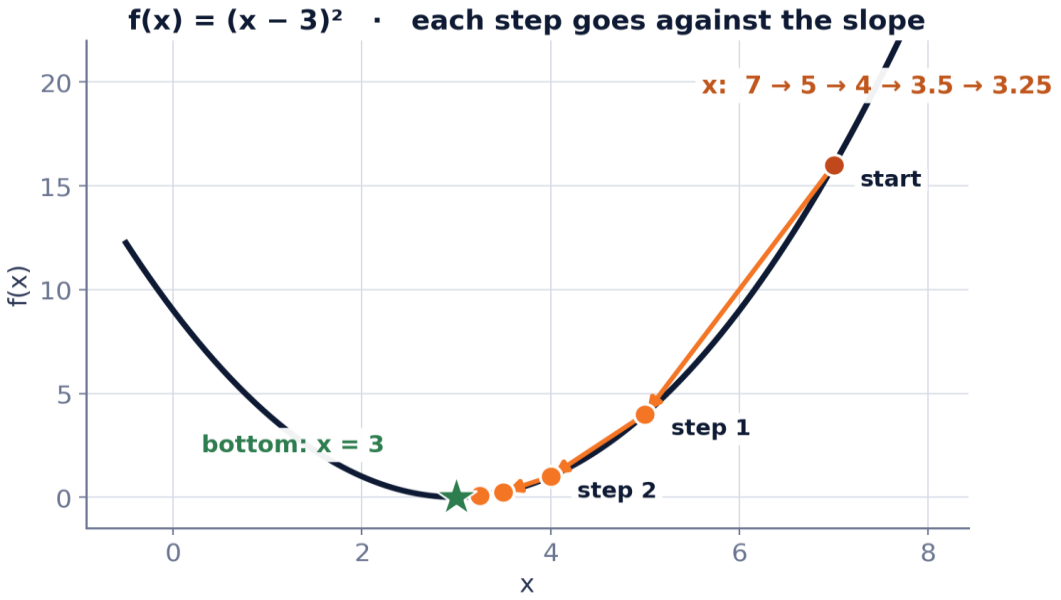
Gradient Descent Is One Line

$$\text{new } x = \text{old } x - \alpha \times \text{slope} \qquad (\alpha = \text{step size, a number you choose})$$

step	x	slope = 2(x – 3)	α × slope (α = 0.25)	new x
0	7	8	2	**5**
1	5	4	1	**4**
2	4	2	0.5	**3.5**
3	3.5	1	0.25	**3.25**

NOTICE

The distance to 3 halves every step: 4 → 2 → 1 → 0.5 → 0.25.



Try It Yourself: $f(x) = (x - 1)^2$

THE TASK

Bottom at $x = 1$. Slope $= 2 \cdot (x - 1)$.
Start at $x = 5$ and use step size $\alpha = 0.1$.
Fill in three steps.

step	x	slope	α × slope	new x
0	5			
1				
2				

BONUS QUESTIONS

A. What happens with $\alpha = 0.5$?

B. What happens with $\alpha = 1$?

C. What happens with $\alpha = 0.01$?

Hint: write the first two steps and look for a pattern.

Answers — and Three Surprises

step	x	slope	α × slope	new x
0	5	8	0.8	**4.2**
1	4.2	6.4	0.64	**3.56**
2	3.56	5.12	0.512	**3.048**

WHAT YOU SEE

Each step removes 20 % of the remaining distance (4 → 3.2 → 2.56 → 2.05). Steady, but slow.

A · α = 0.5

5 → 1. Done in ONE step: the step was exactly right.

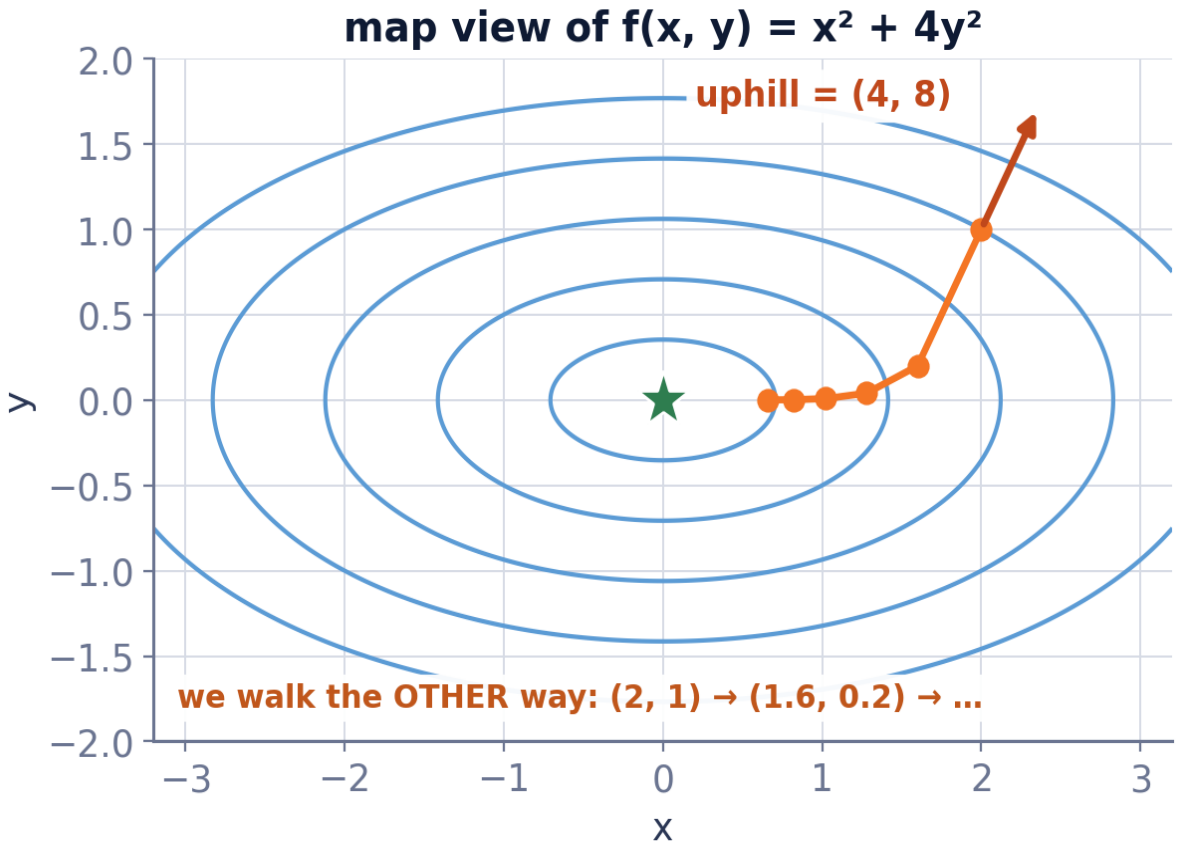
B · α = 1

5 → −3 → 5 → −3 … Bounces forever, never arrives.

C · α = 0.01

5 → 4.92 → 4.84 … Correct direction, but hundreds of steps.

With Two Variables: Read the Map



THE GRADIENT, IN PLAIN WORDS

Measure the slope in each direction separately, and put the numbers side by side. That pair is the gradient.

WORKED EXAMPLE

$f(x, y) = x^2 + 4y^2$ at the point (2, 1)

slope in the x direction: $2x = 4$

slope in the y direction: $8y = 8$

gradient = $(4, 8)$ — points UPHILL

step with $\alpha = 0.1$: $(2, 1) - 0.1 \cdot (4, 8) = (1.6, 0.2)$

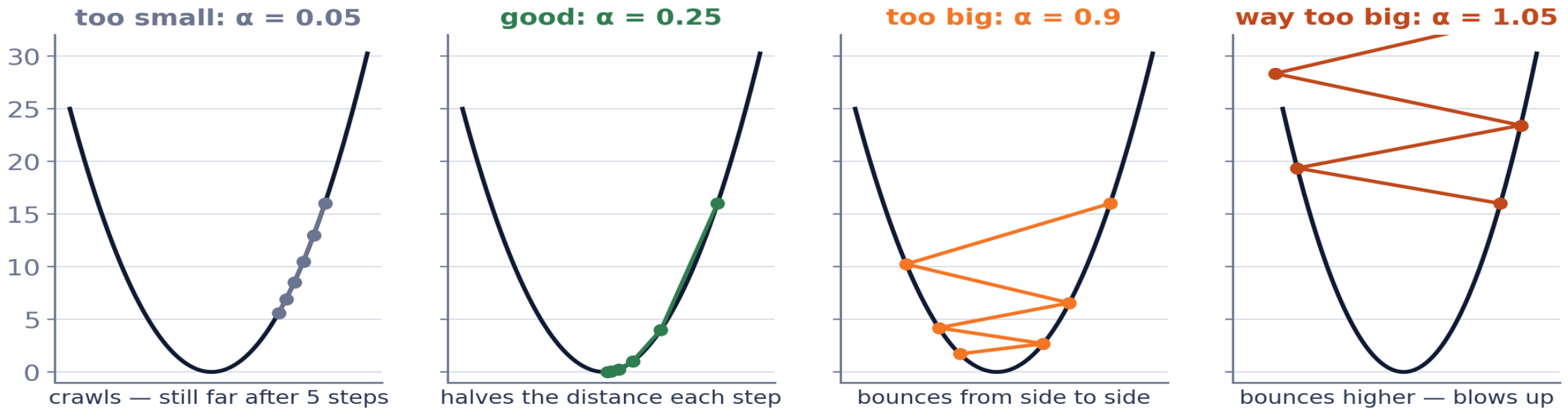
height drops from 8 to 2.72 in one step

PART II

How Big a Step?

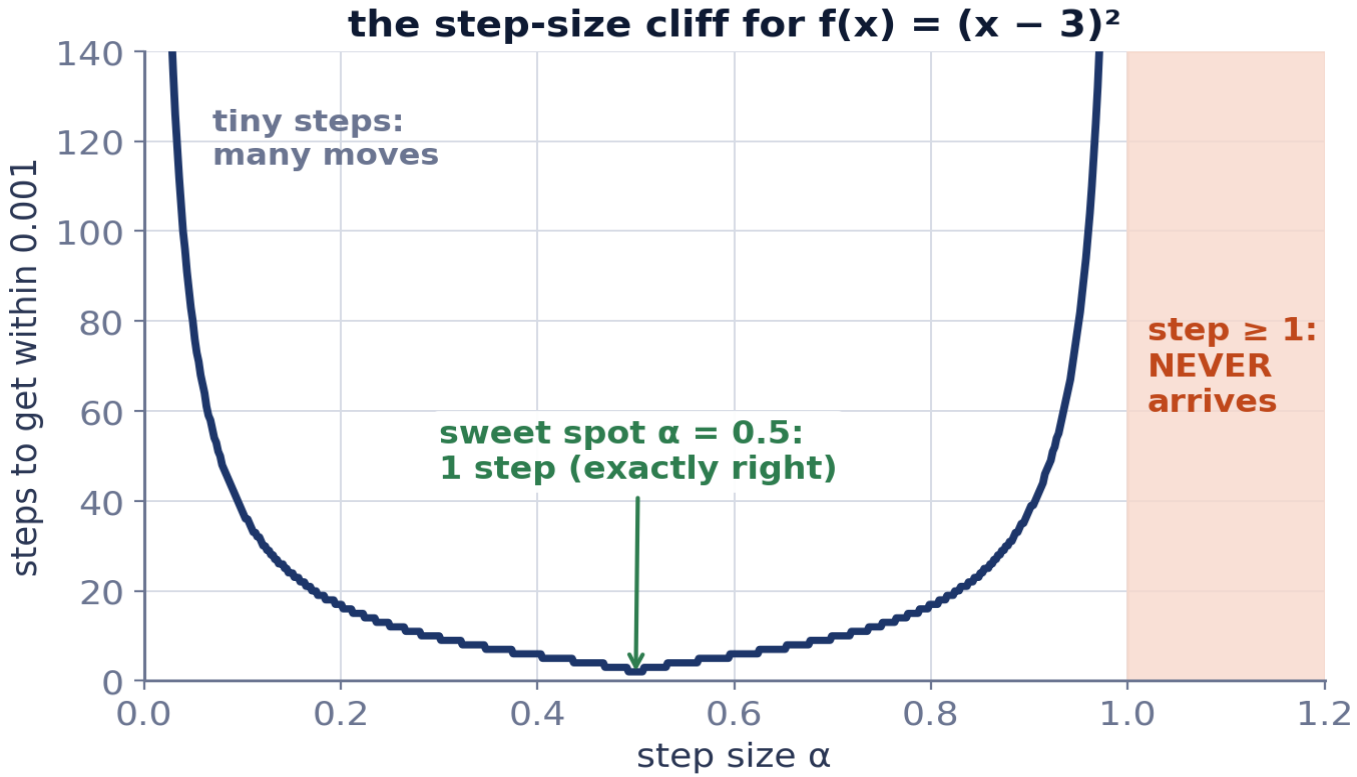
The step size alone can make the same method perfect, slow, or useless.

Too Small, Just Right, Too Big



step size α	x after 5 steps (start 7, target 3)	verdict
0.05	5.36 — still 2.4 away	too slow
0.25	3.13 — almost there	good
0.9	7 → −0.2 → 5.6 → 0.95 → 4.6 → 1.7	bouncing
1.05	7 → −1.4 → 7.8 → −2.3 → 8.9 → −3.4	exploding

There Is a Cliff: Past It, You Never Arrive



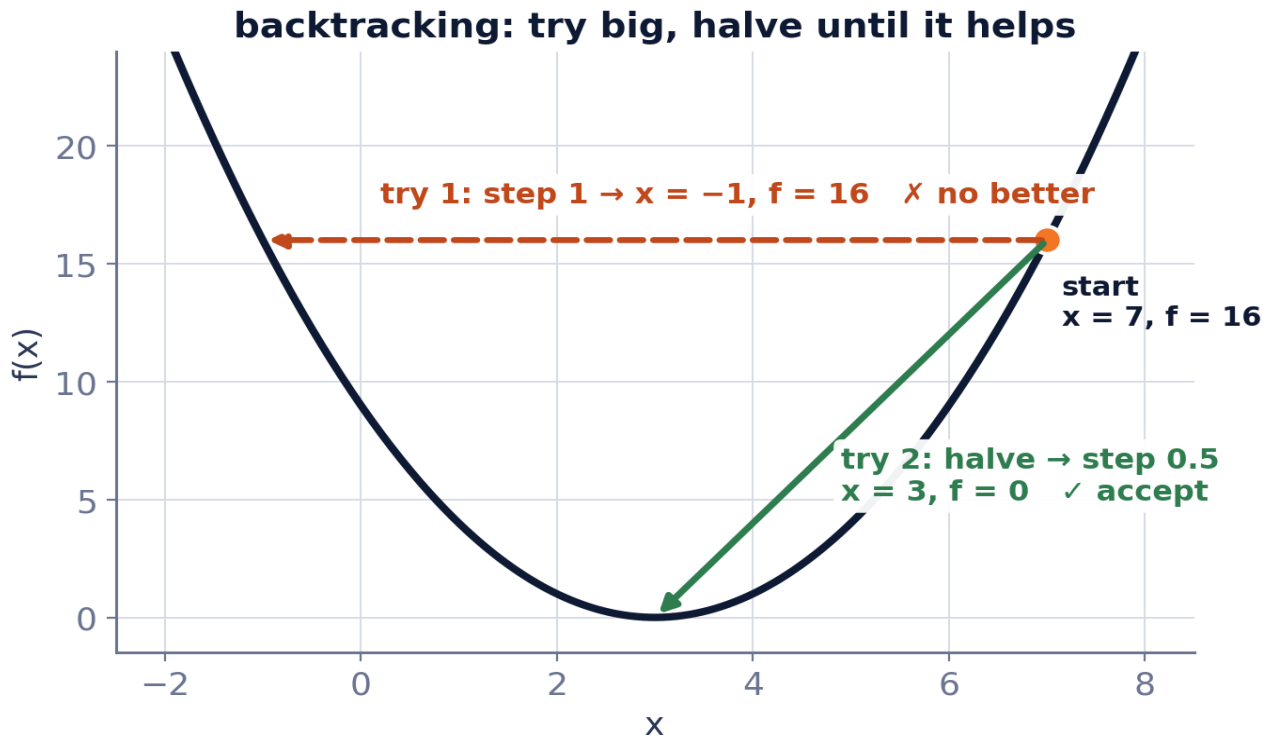
α	steps needed
0.05	79
0.25	12
0.5	1
0.9	38
1.0 or more	never

RULE OF THUMB

The more curved the bowl, the smaller the safe step. Steep, narrow bowls need careful, small steps.

In practice: start with a moderate α. If the error grows or jumps around, make α smaller.

Let the Computer Choose: Try Big, Then Halve



THE RECIPE (BACKTRACKING)

1. Try a big step.
2. Did the height go down? Accept it.
3. If not, halve the step and try again.

```
step = 1.0
while f(x - step*slope) >= f(x):
    step = step / 2      # halve it
x = x - step*slope       # take it
```

OUR EXAMPLE

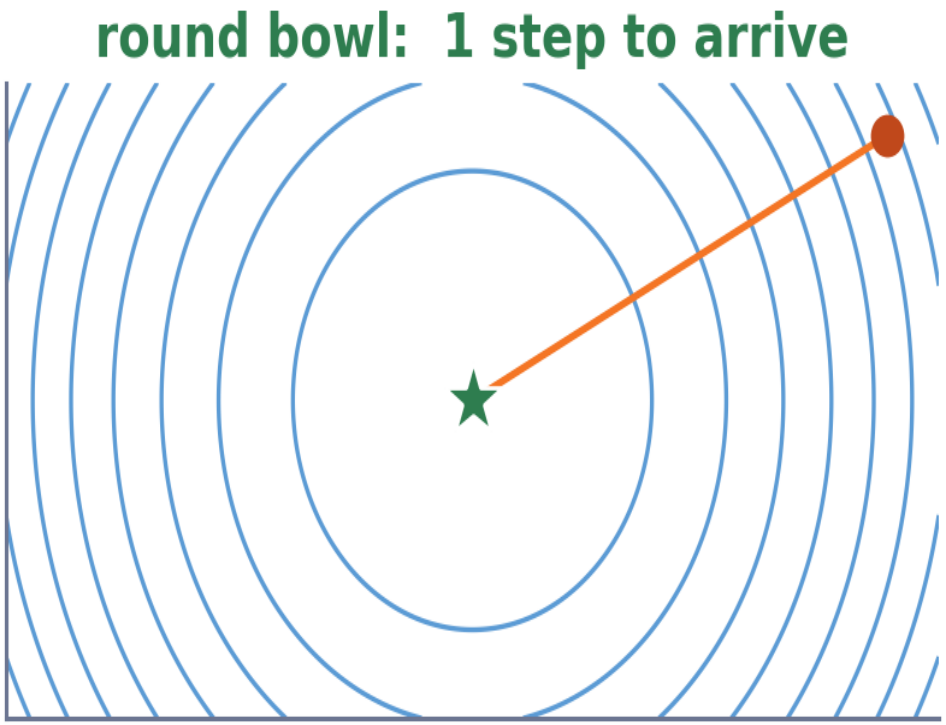
Step 1 lands at the same height (16). Halved to 0.5, it lands on the bottom.

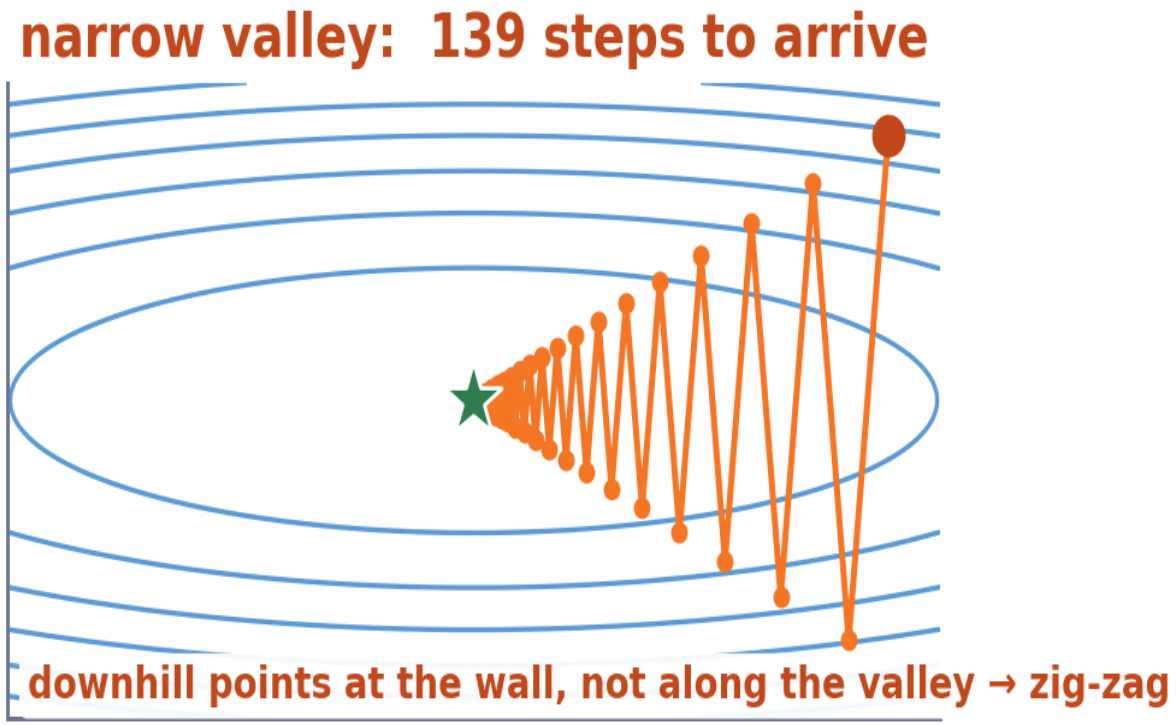
PART III

Why It Can Be Slow — and the One-Line Fix

Narrow valleys make gradient descent zig-zag. Rescaling the data straightens them.

Round Bowls Are Easy, Narrow Valleys Are Slow





ROUND BOWL

Downhill points straight at the bottom → 1 step.

NARROW VALLEY

Downhill points at the steep wall → 139 zig-zag steps.

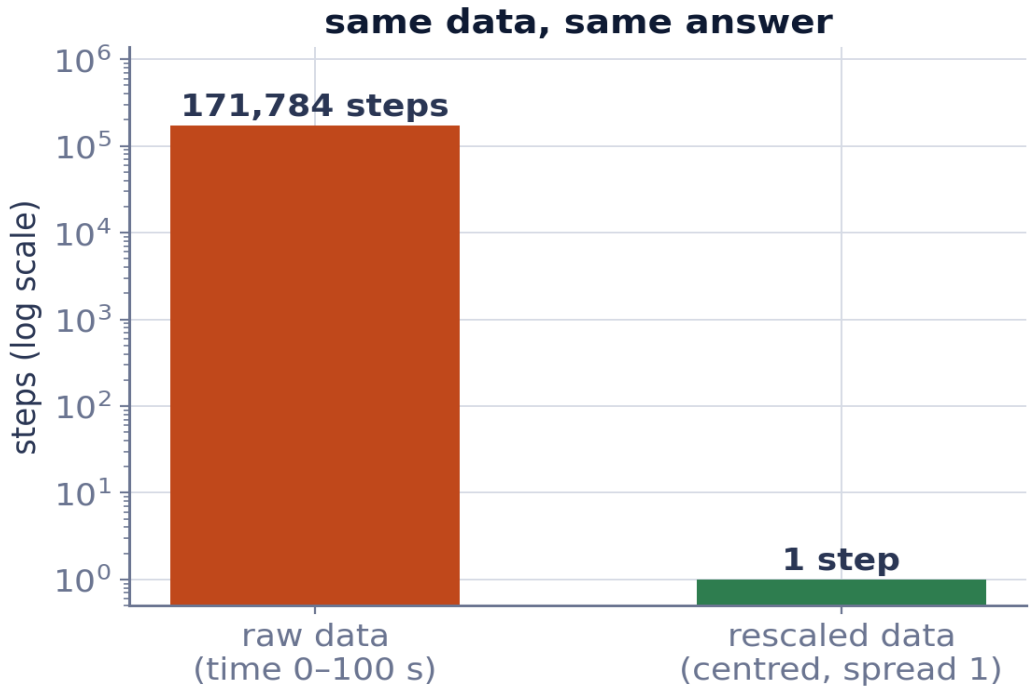
Rescale Your Data: Thousands of Times Faster

WHERE NARROW VALLEYS COME FROM

We fit a line to sensor readings. One input is time (0 to 100 seconds), the other is a constant (always 1). Numbers of very different sizes stretch the bowl into a narrow valley.

THE FIX: STANDARDIZE

For each input: subtract its average, divide by its spread. Now every input has a similar size.




```
t_scaled = (t - t.mean()) / t.std()
```

Raw data: 171,784 steps. Rescaled: 1 step. Same final line: y = 2.84 + 0.487·t

Three Practical Stopping Rules

1

The ground is flat

Stop when the slope is almost zero, e.g. smaller than 0.000001.

In our table the slope went 8 → 4 → 2 → 1 → … → nearly 0.

2

Progress is tiny

Stop when one step improves the error by almost nothing.

Heights: 16 → 4 → 1 → 0.25 → 0.06 — the improvements shrink.

3

Out of budget

Always set a maximum number of steps as a safety net.

Protects you from a step size that bounces forever.

Real code uses all three together: stop when ANY of them is true.

Always Check Your Slope Formula: the Nudge Test

THE IDEA

Nudge x a tiny bit to the right and to the left, measure how much f changes, and divide by the distance. That estimates the slope without any formula.

WORKED EXAMPLE

Check the slope of $(x - 3)^2$ at x = 7

$f(7.001) = 16.008001 \qquad f(6.999) = 15.992001$

$(16.008001 - 15.992001) / 0.002 = 8.000$

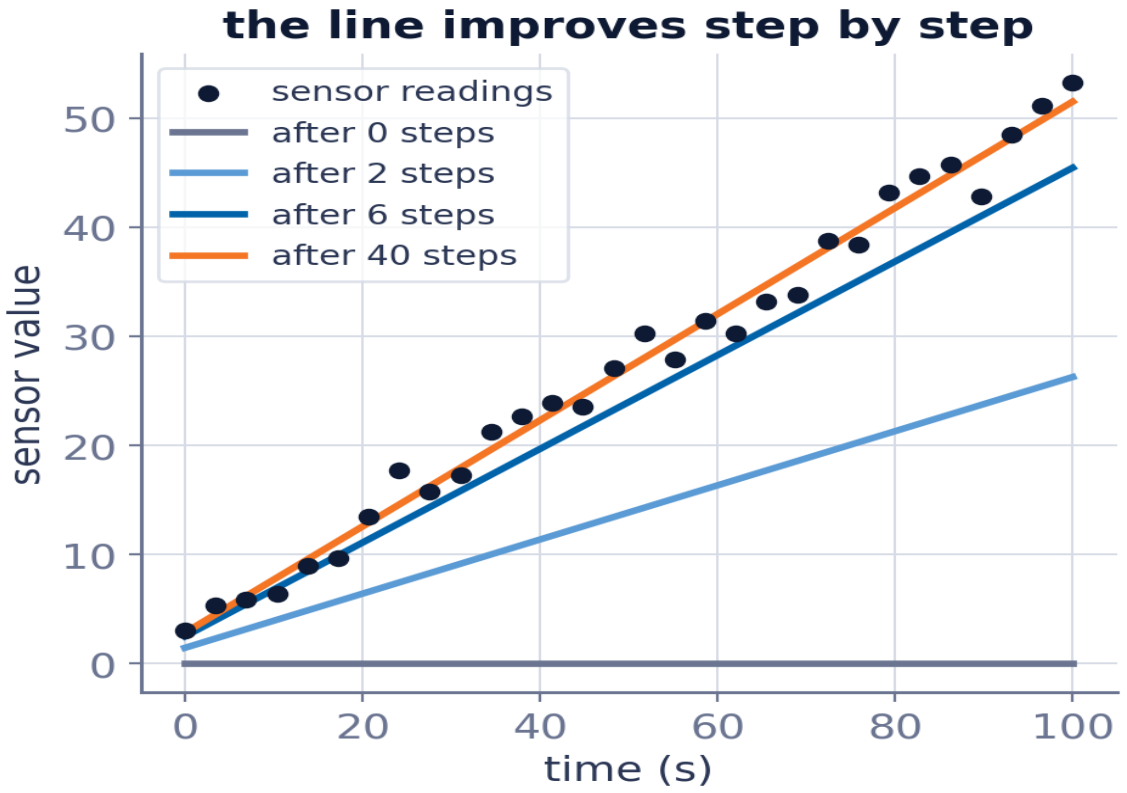
formula $2 \cdot (x - 3) = 8 \quad \rightarrow \quad$ they agree ✓

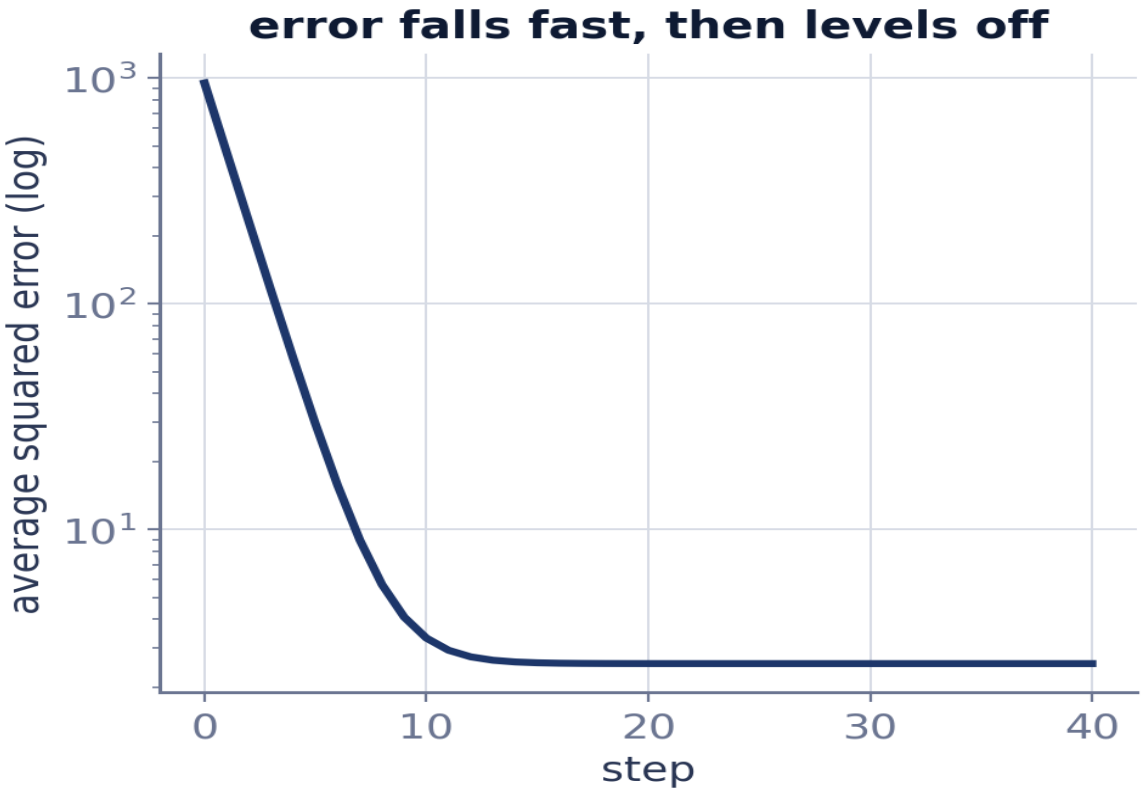
WHY BOTHER?

A wrong slope formula does not crash your program. Gradient descent just walks to the wrong place, or very slowly — and you may never notice.

Example bug: you typed slope = x − 3 (forgot the 2).
The nudge test says 8, your formula says 4.
Bug found in one second.

Watch It Work: Fitting a Line to Sensor Data





step	0	2	6	40
average squared error	951.8	230.5	15.7	**2.55**

Where You Will Meet Gradient Descent

Training neural networks

Every image, speech and language model is trained by a version of this loop (Weeks 10–11).

Fitting models to data

Regression, calibration, curve fitting — as in today's sensor example.

Recommendation systems

Finding user and item preferences that best explain past ratings.

Engineering design

Tuning controller gains, antenna shapes, circuit parameters by simulation.

Next week: smarter steps — using the CURVE of the bowl, not just the slope, to cross narrow valleys fast.

Lab 5: Ten Lines of Python

```
def f(x):     return (x - 3)**2
def slope(x): return 2*(x - 3)

x, alpha = 7.0, 0.25
for step in range(100):
    g = slope(x)
    if abs(g) < 1e-6:          # flat: stop
        break
    x = x - alpha * g          # the one line
print(step, x)
```

WHAT YOU WILL DO

1. Run the code. Reproduce today's table.
2. Change α: find the cliff yourself.
3. Add backtracking (4 more lines).
4. Fit the sensor line — raw vs rescaled data. Count the steps.

The full lab notebook (Week05_Lab.ipynb) has the details; start with sections 1–3.

Six Rules of Thumb

1. **Go against the slope.**
   new x = x − α × slope

2. **The step size decides everything.**
   too small crawls, too big bounces or explodes

3. **If the error grows, shrink α.**
   or let backtracking choose it for you

4. **Narrow valleys cause zig-zag.**
   the steep direction limits the step

5. **Rescale your data first.**
   171,784 steps became 1

6. **Check your slope, set a step budget.**
   the nudge test catches silent bugs

WEEK 5 — TAKEAWAYS

1. **One line —** new x = x − α × slope. That is gradient descent.

2. **One example —** $(x - 3)^2$ from x = 7: 7 → 5 → 4 → 3.5 → 3.25.

3. **One danger —** a step that is too big bounces or explodes.

4. **One fix —** rescale the data: narrow valleys become round bowls.